Data Communications and Networking II Project
Wireshark Lab: IP

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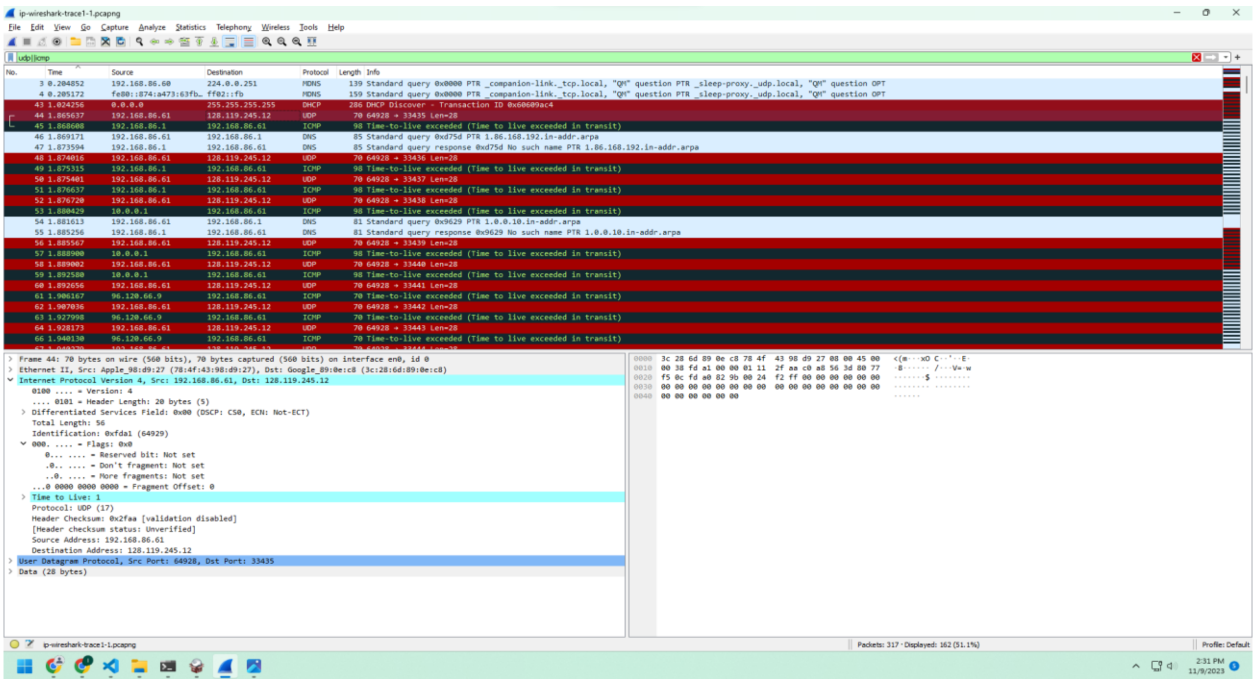
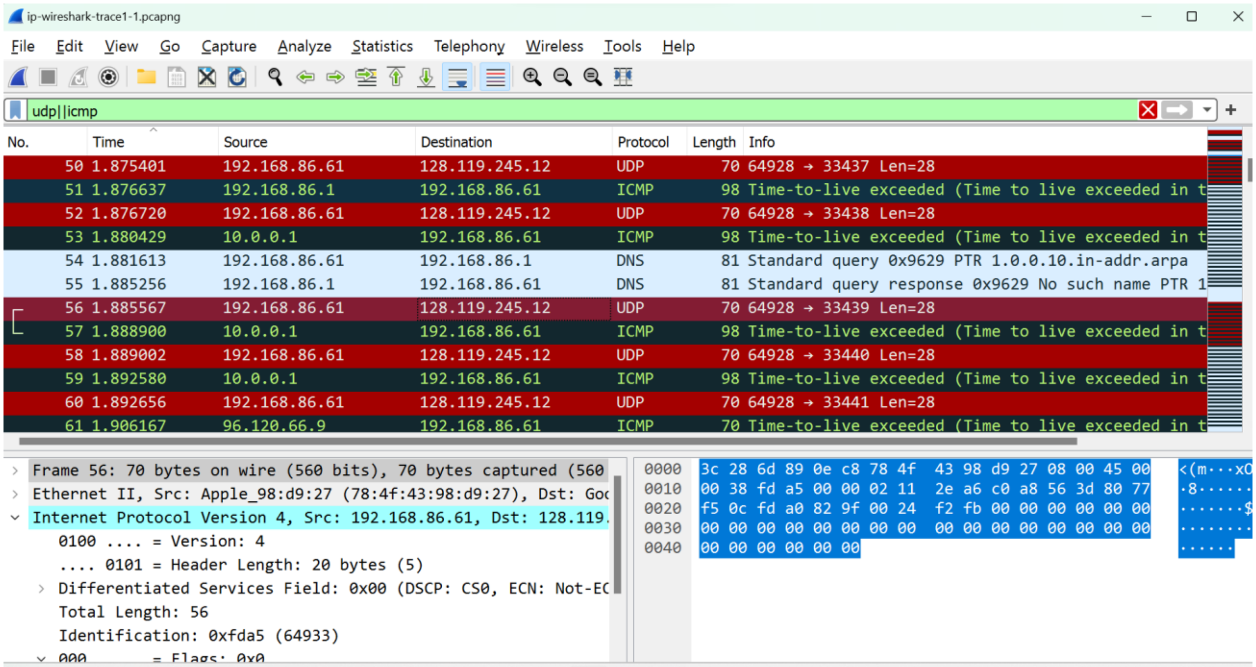
Student ID:801366046

PART I





1. Select the first UDP segment sent by your computer via the `traceroute` command to gaia.cs.umass.edu. (Hint: this is 44th packet in the trace file in the *ipwireshark-trace1-1.pcapng* file). Expand the Internet Protocol part of the packet in the packet details window. What is the IP address of your computer?
   <u>ANSWER:</u>
   The IP of my computer is 192.168.86.61[IPV4].

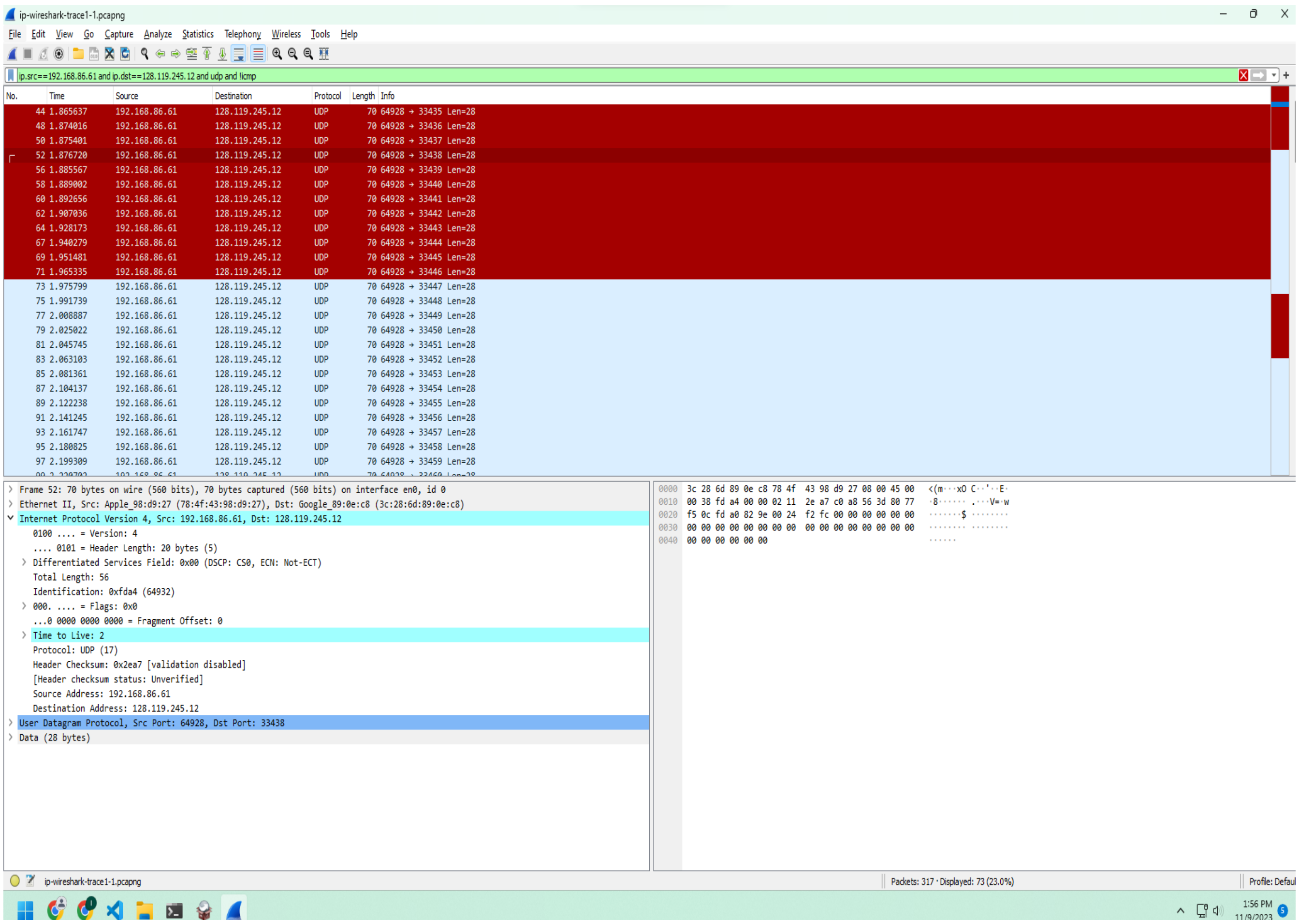
2. What is the value in the time-to-live (TTL) field in this IPv4 datagram's header?
   <u>ANSWER:</u>
   TTL: 1

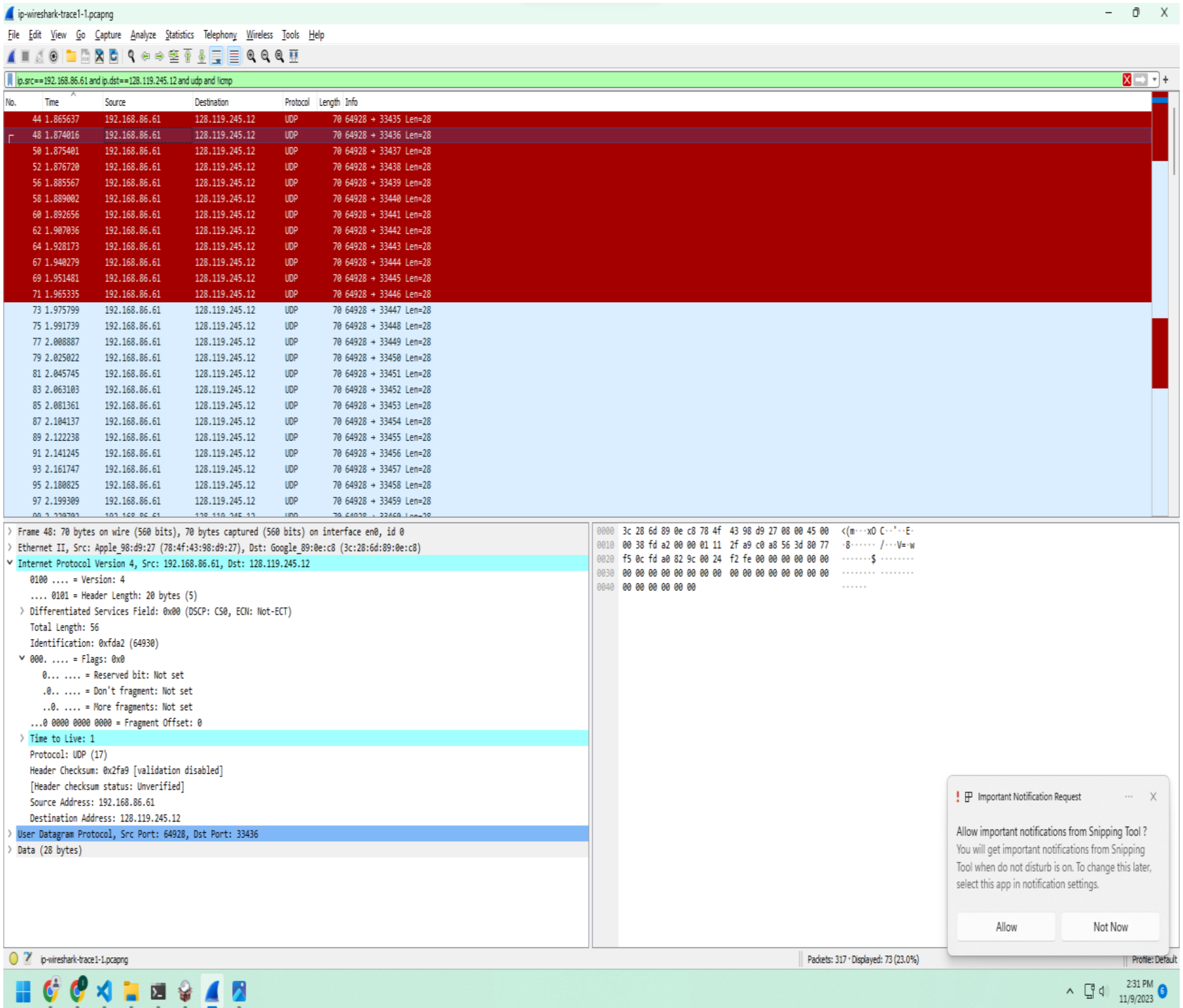
3. What is the value in the upper layer protocol field in this IPv4 datagram's header?
   <u>ANSWER:</u>
   Upper layer protocol field in IPV4 datagram's header: UDP(17).

4. How many bytes are in the IP header?
   <u>ANSWER:</u>
   IP Header length: 20bytes.

5. How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.
   <u>ANSWER:</u>
   Total Length = 56
   Data Length = Total length – Header length
   = 56-20
   = 36

6. Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.
   <u>ANSWER:</u>
   Since more fragment and don’t fragment are not set, the datagram has not been fragmented.





7. Which fields in the IP datagram *always* change from one datagram to the next within this series of UDP segments sent by your computer destined to 128.119.245.12, via `traceroute`? Why? (List ALL the fields that change)

   <u>ANSWER:</u>

   Fields in IP datagram always change from one datagram to the next datagram within series of UDP segments:

   - <u>Time To Live:</u>
     In received datagram, router decrements TTL by 1.

   - <u>Identification:</u>
     It should be unique because it is required for reassembly.

   - <u>Header Checksum:</u>
     It changes since it is reverified & recomputed at each address.

8. Which fields in this sequence of IP datagrams (containing UDP segments) stay constant? Why? (List ALL the fields that stay constant)

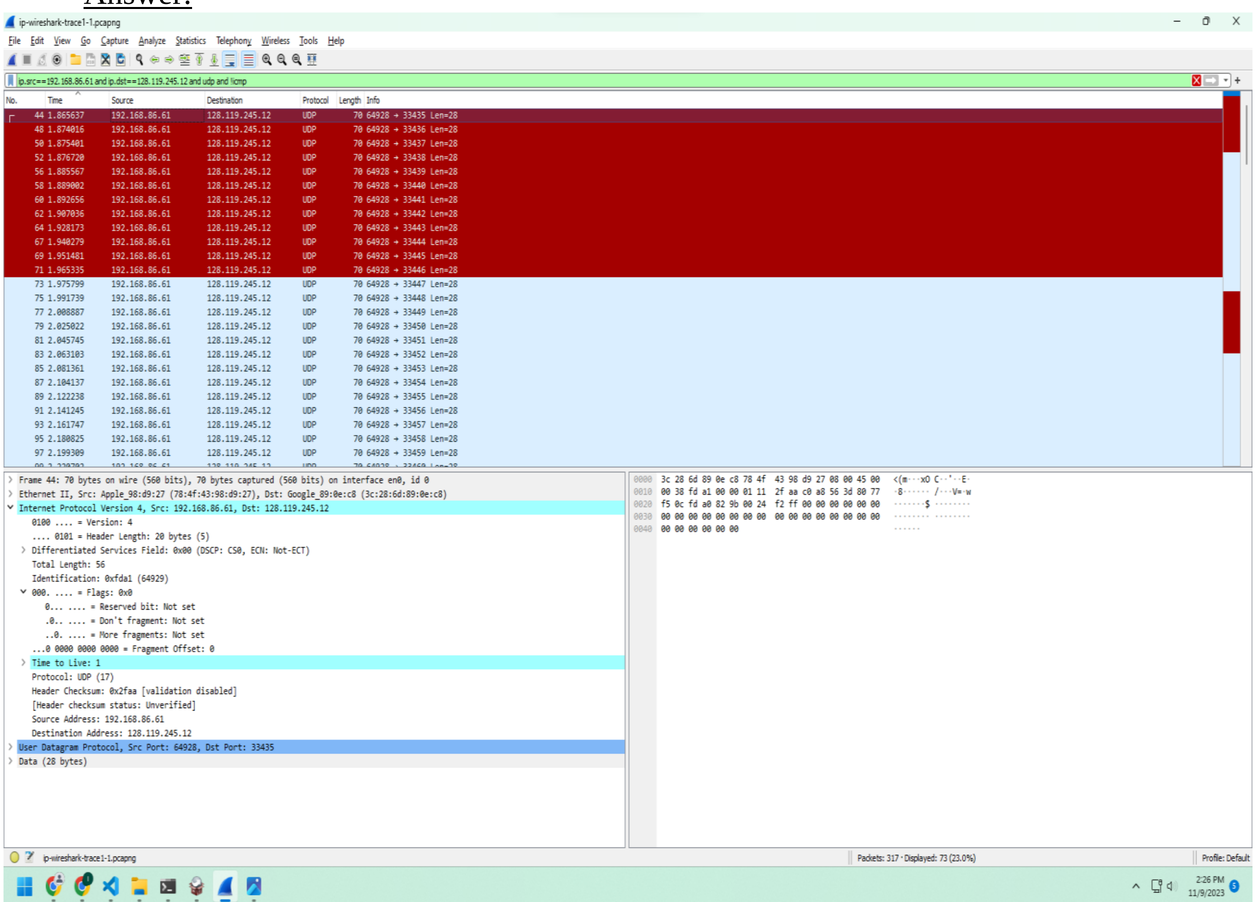
   Answer:

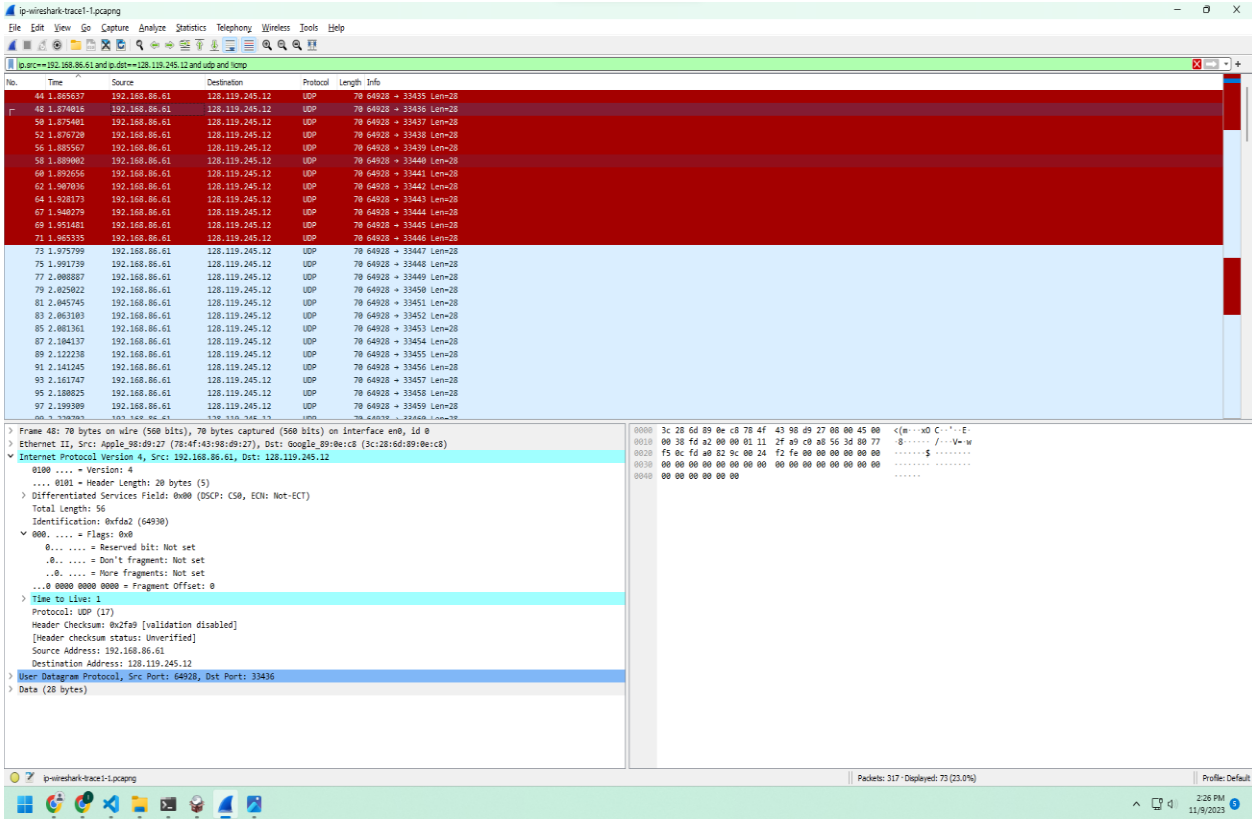
   Fields in IP datagram stay constant:

   - Version: Since, here using IPV4
   - Header length: since all packets are UDP
   - Differentiated Service: Since, all packets are UDP
   - Protocol: Since all packets are UDP
   - Source address: Since, all packets are sent from my computer
   - Destination address: Since, all packets are sent to same host
   - Flags : The Don't fragment and more fragment bits are not set because the packets are not fragmented here(as attached in screenshots below).

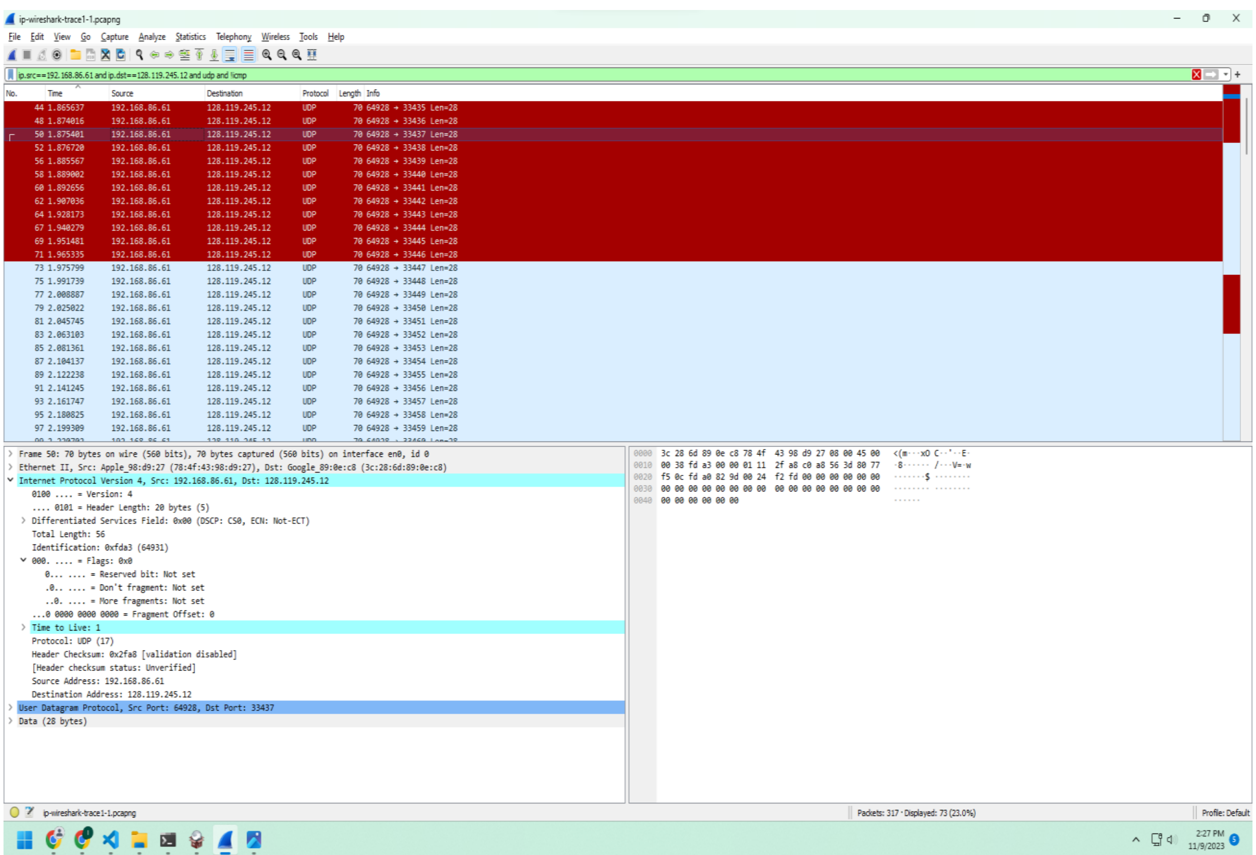
9. Describe the pattern you see in the values in the Identification field of the IP datagrams being sent by your computer.

   Answer:

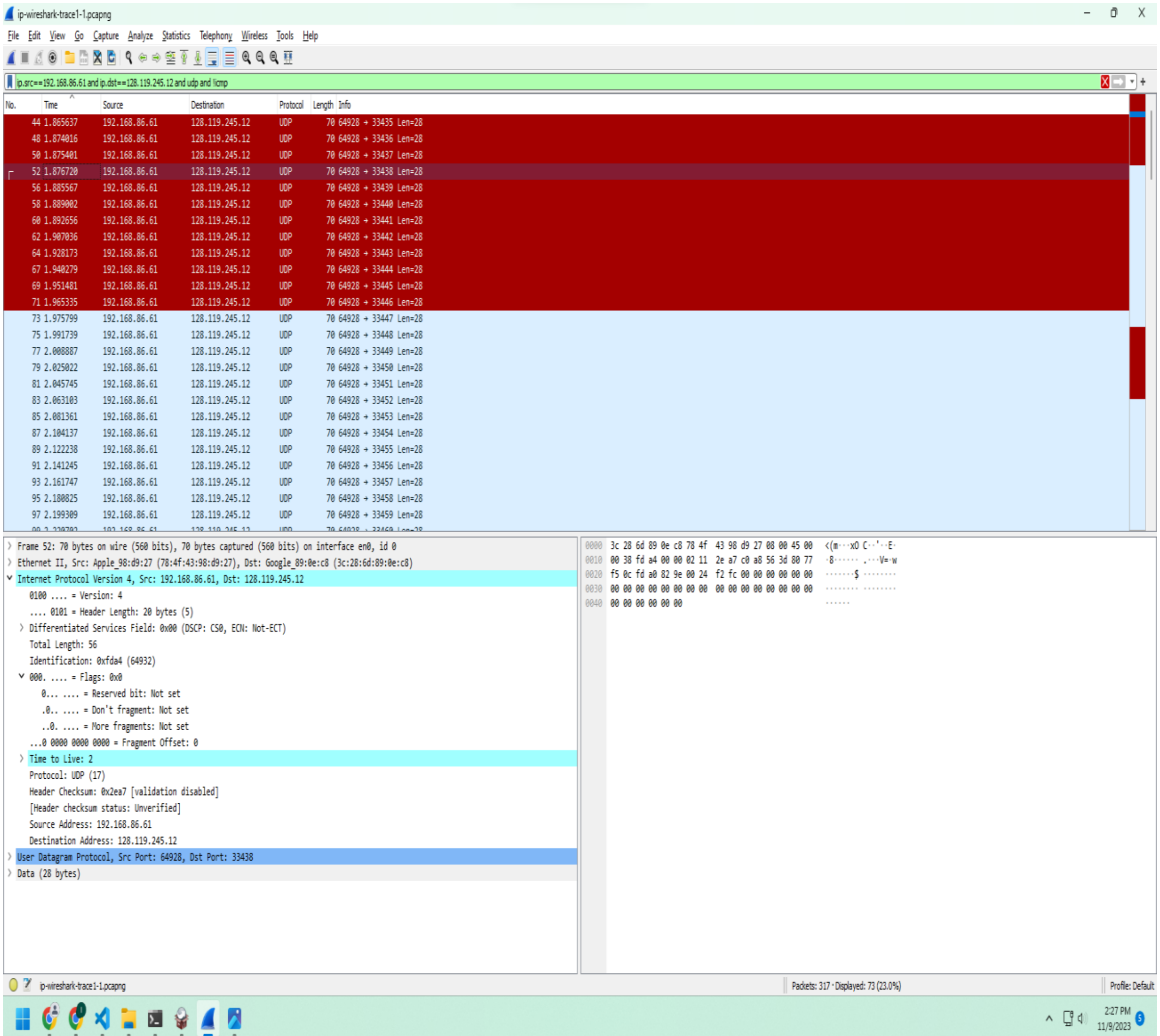




ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and !icmp
No. Time Source Destination Protocol Length Info
44 1.865637 192.168.86.61 128.119.245.12 UDP 70 64928 → 33435 Len=28
48 1.874016 192.168.86.61 128.119.245.12 UDP 70 64928 → 33436 Len=28
50 1.875401 192.168.86.61 128.119.245.12 UDP 70 64928 → 33437 Len=28
52 1.876720 192.168.86.61 128.119.245.12 UDP 70 64928 → 33438 Len=28
56 1.885567 192.168.86.61 128.119.245.12 UDP 70 64928 → 33439 Len=28
58 1.889002 192.168.86.61 128.119.245.12 UDP 70 64928 → 33440 Len=28
60 1.892656 192.168.86.61 128.119.245.12 UDP 70 64928 → 33441 Len=28
62 1.907036 192.168.86.61 128.119.245.12 UDP 70 64928 → 33442 Len=28
64 1.928173 192.168.86.61 128.119.245.12 UDP 70 64928 → 33443 Len=28
67 1.940279 192.168.86.61 128.119.245.12 UDP 70 64928 → 33444 Len=28
69 1.951481 192.168.86.61 128.119.245.12 UDP 70 64928 → 33445 Len=28
71 1.965335 192.168.86.61 128.119.245.12 UDP 70 64928 → 33446 Len=28
73 1.975799 192.168.86.61 128.119.245.12 UDP 70 64928 → 33447 Len=28
75 1.991739 192.168.86.61 128.119.245.12 UDP 70 64928 → 33448 Len=28
77 2.008887 192.168.86.61 128.119.245.12 UDP 70 64928 → 33449 Len=28
79 2.025022 192.168.86.61 128.119.245.12 UDP 70 64928 → 33450 Len=28
81 2.045745 192.168.86.61 128.119.245.12 UDP 70 64928 → 33451 Len=28
83 2.063103 192.168.86.61 128.119.245.12 UDP 70 64928 → 33452 Len=28
85 2.081361 192.168.86.61 128.119.245.12 UDP 70 64928 → 33453 Len=28
87 2.104137 192.168.86.61 128.119.245.12 UDP 70 64928 → 33454 Len=28
89 2.122238 192.168.86.61 128.119.245.12 UDP 70 64928 → 33455 Len=28
91 2.141245 192.168.86.61 128.119.245.12 UDP 70 64928 → 33456 Len=28
93 2.161747 192.168.86.61 128.119.245.12 UDP 70 64928 → 33457 Len=28
95 2.180825 192.168.86.61 128.119.245.12 UDP 70 64928 → 33458 Len=28
97 2.199309 192.168.86.61 128.119.245.12 UDP 70 64928 → 33459 Len=28
Frame 48: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda2 (64930)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x2fa9 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33436
Data (28 bytes)
0000 3c 28 6d 89 0e c8 78 4f 43 98 d9 27 08 00 45 00
0010 00 38 fd a2 00 00 01 11 2f a9 c0 a8 56 3d 80 77
0020 f5 0c fd a0 82 9c 00 24 f2 fe 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00
ip-wireshark-trace1-1.pcapng
Packets: 317 · Displayed: 73 (23.0%)
Profile: Default
2:26 PM
11/9/2023



ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and !icmp
No. Time Source Destination Protocol Length Info
44 1.865637 192.168.86.61 128.119.245.12 UDP 70 64928 → 33435 Len=28
48 1.874016 192.168.86.61 128.119.245.12 UDP 70 64928 → 33436 Len=28
50 1.875401 192.168.86.61 128.119.245.12 UDP 70 64928 → 33437 Len=28
52 1.876720 192.168.86.61 128.119.245.12 UDP 70 64928 → 33438 Len=28
56 1.885567 192.168.86.61 128.119.245.12 UDP 70 64928 → 33439 Len=28
58 1.889002 192.168.86.61 128.119.245.12 UDP 70 64928 → 33440 Len=28
60 1.892656 192.168.86.61 128.119.245.12 UDP 70 64928 → 33441 Len=28
62 1.907036 192.168.86.61 128.119.245.12 UDP 70 64928 → 33442 Len=28
64 1.928173 192.168.86.61 128.119.245.12 UDP 70 64928 → 33443 Len=28
67 1.940279 192.168.86.61 128.119.245.12 UDP 70 64928 → 33444 Len=28
69 1.951481 192.168.86.61 128.119.245.12 UDP 70 64928 → 33445 Len=28
71 1.965335 192.168.86.61 128.119.245.12 UDP 70 64928 → 33446 Len=28
73 1.975799 192.168.86.61 128.119.245.12 UDP 70 64928 → 33447 Len=28
75 1.991739 192.168.86.61 128.119.245.12 UDP 70 64928 → 33448 Len=28
77 2.008887 192.168.86.61 128.119.245.12 UDP 70 64928 → 33449 Len=28
79 2.025022 192.168.86.61 128.119.245.12 UDP 70 64928 → 33450 Len=28
81 2.045745 192.168.86.61 128.119.245.12 UDP 70 64928 → 33451 Len=28
83 2.063103 192.168.86.61 128.119.245.12 UDP 70 64928 → 33452 Len=28
85 2.081361 192.168.86.61 128.119.245.12 UDP 70 64928 → 33453 Len=28
87 2.104137 192.168.86.61 128.119.245.12 UDP 70 64928 → 33454 Len=28
89 2.122238 192.168.86.61 128.119.245.12 UDP 70 64928 → 33455 Len=28
91 2.141245 192.168.86.61 128.119.245.12 UDP 70 64928 → 33456 Len=28
93 2.161747 192.168.86.61 128.119.245.12 UDP 70 64928 → 33457 Len=28
95 2.180825 192.168.86.61 128.119.245.12 UDP 70 64928 → 33458 Len=28
97 2.199309 192.168.86.61 128.119.245.12 UDP 70 64928 → 33459 Len=28
Frame 50: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda3 (64931)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x2fa8 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33437
Data (28 bytes)
0000 3c 28 6d 89 0e c8 78 4f 43 98 d9 27 08 00 45 00
0010 00 38 fd a3 00 00 01 11 2f a8 c0 a8 56 3d 80 77
0020 f5 0c fd a0 82 9d 00 24 f2 fd 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00
ip-wireshark-trace1-1.pcapng
Packets: 317 · Displayed: 73 (23.0%)
Profile: Default
2:27 PM
11/9/2023


ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and !icmp
No. Time Source Destination Protocol Length Info
44 1.865637 192.168.86.61 128.119.245.12 UDP 70 64928 → 33435 Len=28
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50 1.875401 192.168.86.61 128.119.245.12 UDP 70 64928 → 33437 Len=28
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69 1.951481 192.168.86.61 128.119.245.12 UDP 70 64928 → 33445 Len=28
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73 1.975799 192.168.86.61 128.119.245.12 UDP 70 64928 → 33447 Len=28
75 1.991739 192.168.86.61 128.119.245.12 UDP 70 64928 → 33448 Len=28
77 2.008887 192.168.86.61 128.119.245.12 UDP 70 64928 → 33449 Len=28
79 2.025022 192.168.86.61 128.119.245.12 UDP 70 64928 → 33450 Len=28
81 2.045745 192.168.86.61 128.119.245.12 UDP 70 64928 → 33451 Len=28
83 2.063103 192.168.86.61 128.119.245.12 UDP 70 64928 → 33452 Len=28
85 2.081361 192.168.86.61 128.119.245.12 UDP 70 64928 → 33453 Len=28
87 2.104137 192.168.86.61 128.119.245.12 UDP 70 64928 → 33454 Len=28
89 2.122238 192.168.86.61 128.119.245.12 UDP 70 64928 → 33455 Len=28
91 2.141245 192.168.86.61 128.119.245.12 UDP 70 64928 → 33456 Len=28
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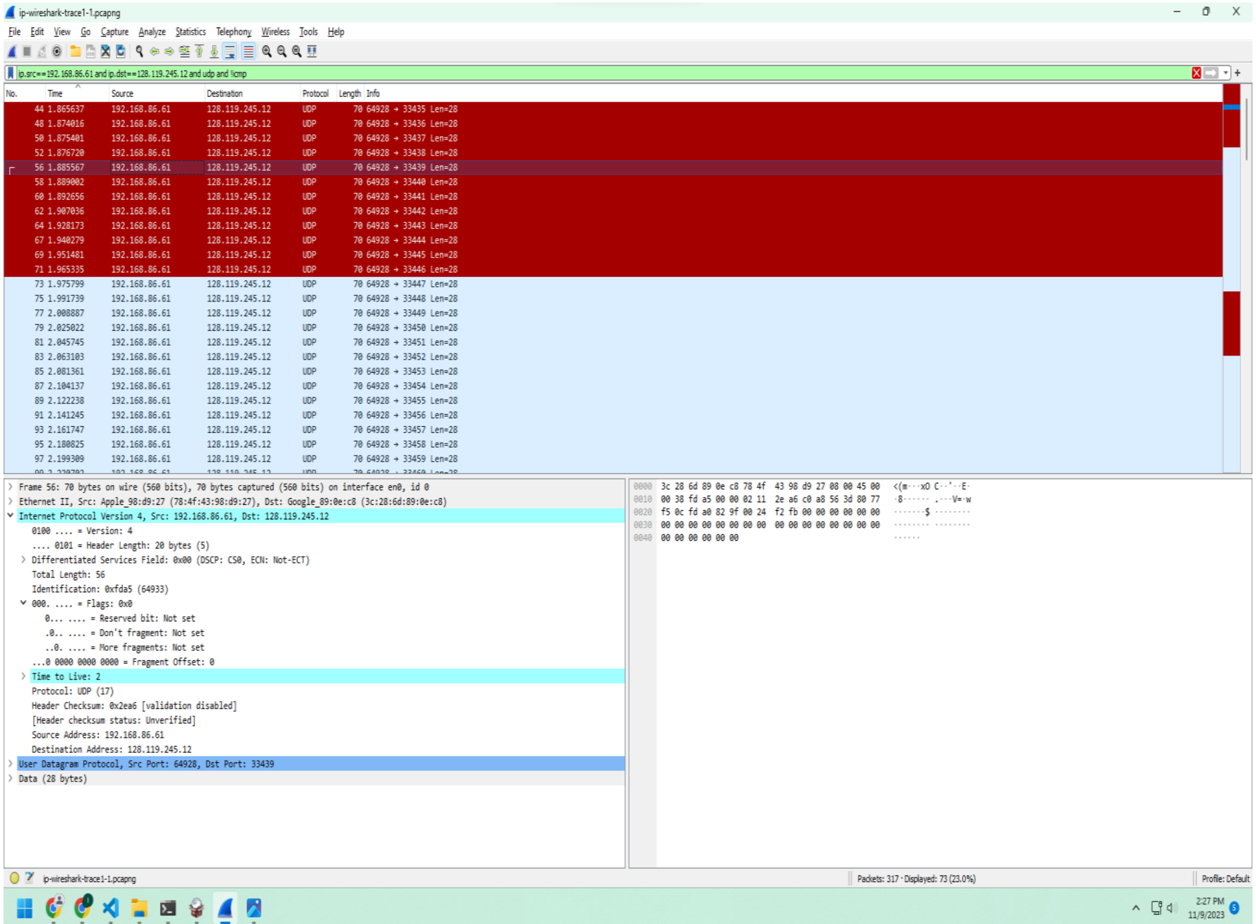
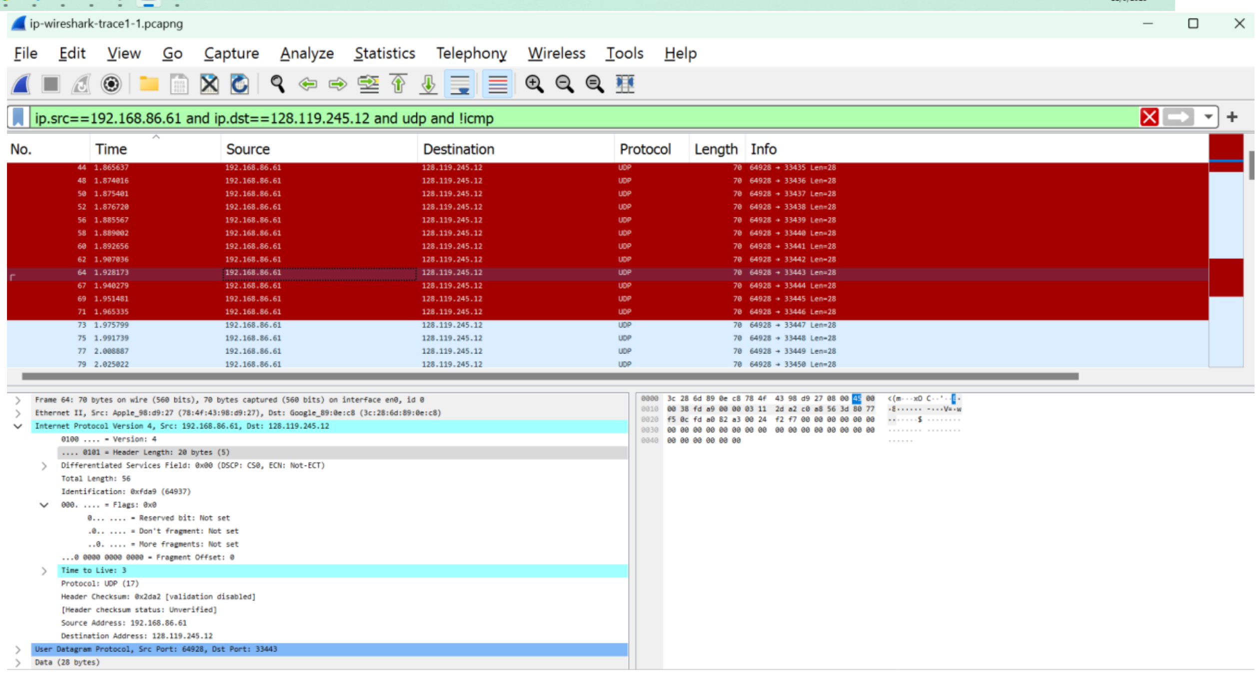
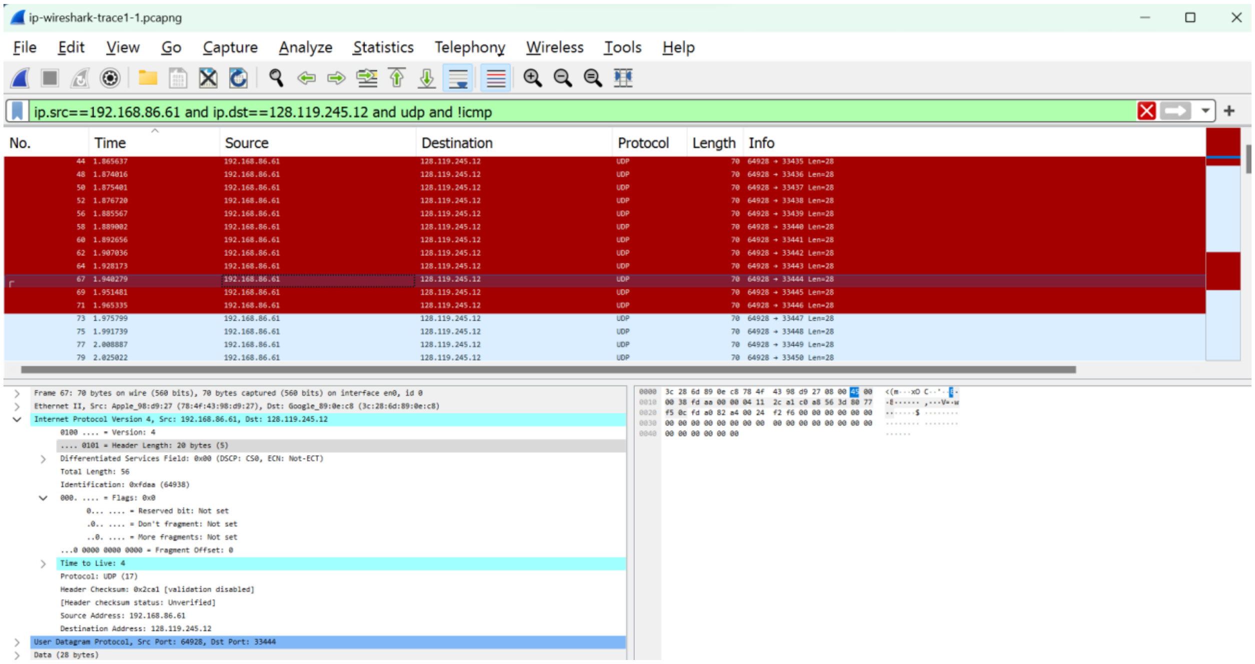
95 2.180825 192.168.86.61 128.119.245.12 UDP 70 64928 → 33458 Len=28
97 2.199309 192.168.86.61 128.119.245.12 UDP 70 64928 → 33459 Len=28
Frame 52: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda4 (64932)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 2
Protocol: UDP (17)
Header Checksum: 0x2ea7 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33438
Data (28 bytes)
0000 3c 28 6d 89 0e c8 78 4f 43 98 d9 27 08 00 45 00
0010 00 38 fd a4 00 00 02 11 2e a7 c0 a8 56 3d 80 77
0020 f5 0c fd a0 82 9e 00 24 f2 fc 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00
ip-wireshark-trace1-1.pcapng
Packets: 317 · Displayed: 73 (23.0%)
Profile: Default
2:27 PM
11/9/2023


ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and !icmp
No. Time Source Destination Protocol Length Info
Frame 56: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda5 (64933)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 2
Protocol: UDP (17)
Header Checksum: 0x2ea6 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33439
Data (28 bytes)
Packets: 317 · Displayed: 73 (23.0%)
Profile: Default
2:27 PM
11/9/2023


ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and !icmp
No. Time Source Destination Protocol Length Info
Frame 64: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda9 (64937)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 3
Protocol: UDP (17)
Header Checksum: 0x2da2 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33443
Data (28 bytes)

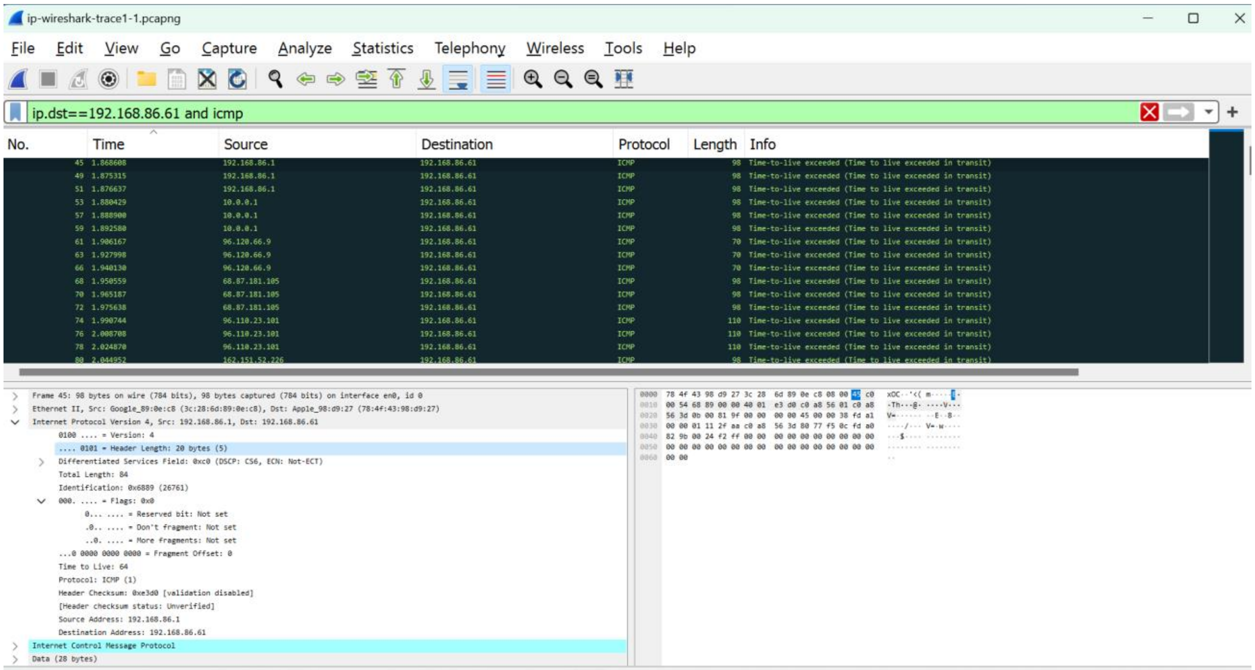


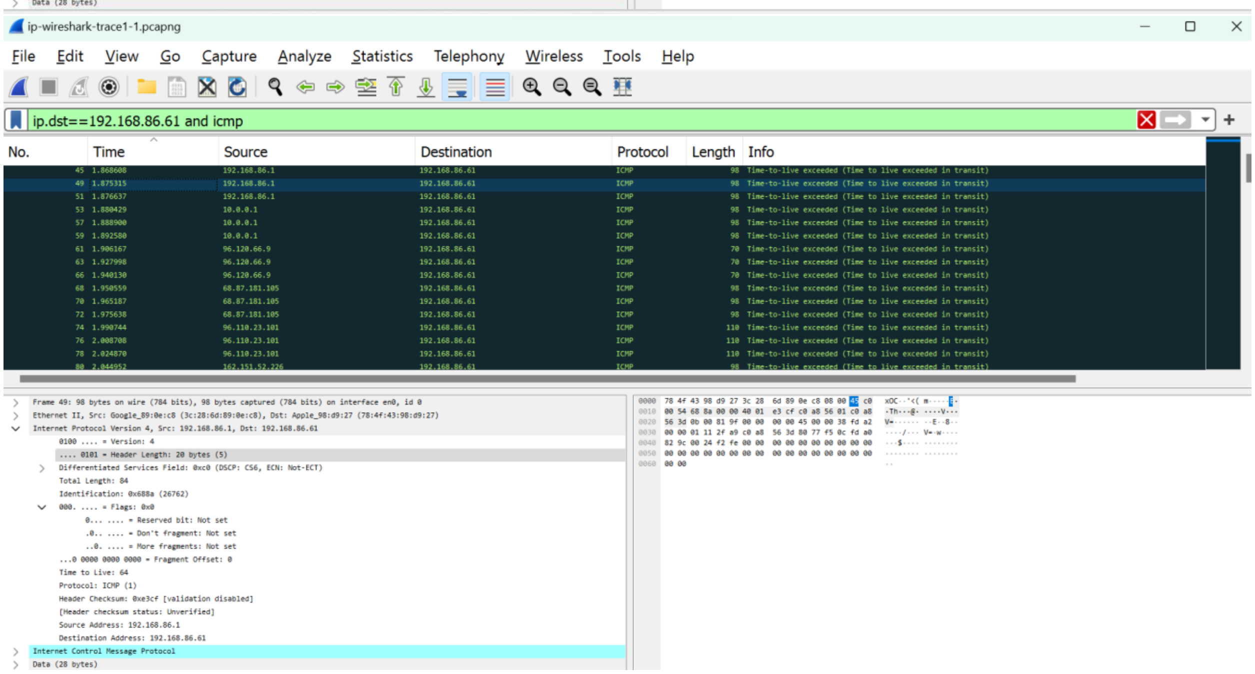
The pattern in the values in the Identification field of the IP datagrams being sent by my computer changes in Hexadecimal format (0 to f) and increases by 1 after each packet.

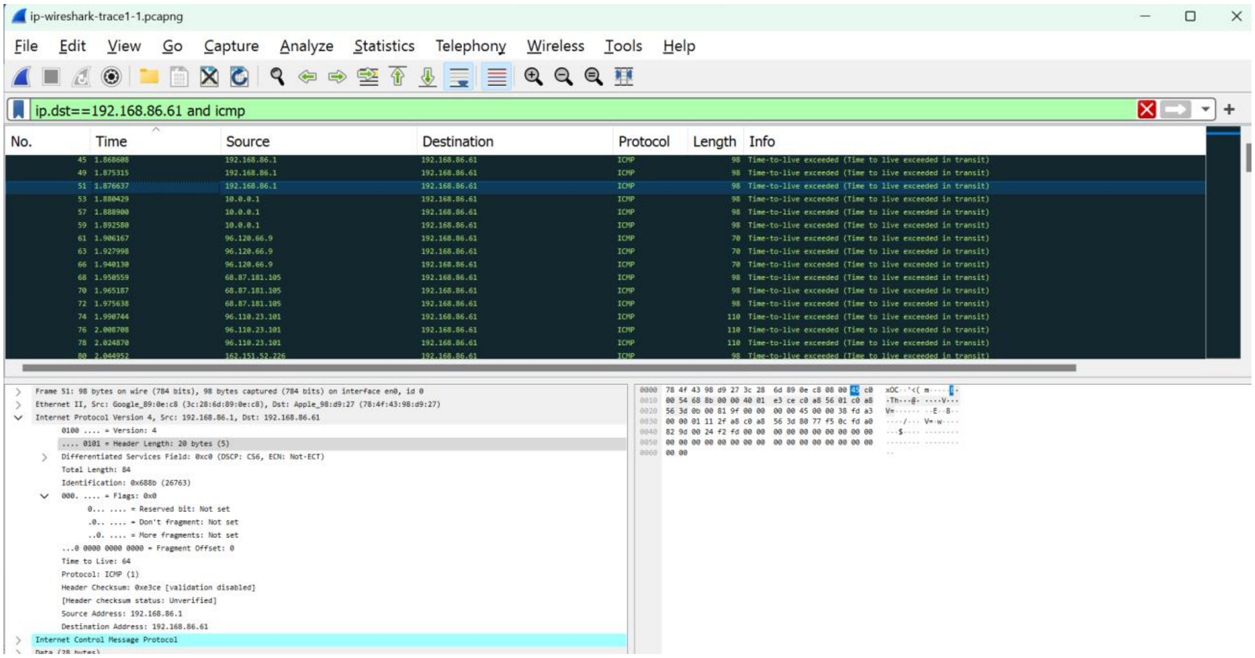
10. What is the upper layer protocol specified in the IP datagrams returned from the routers?
ANSWER:
Upper Layer Protocol: ICMP(1)

11. Are the values in the Identification fields (across the sequence of all of ICMP packets from all of the routers) similar in behavior to your answer to question 9 above?
ANSWER:
No, the values in the identification fields are not similar to question 9.

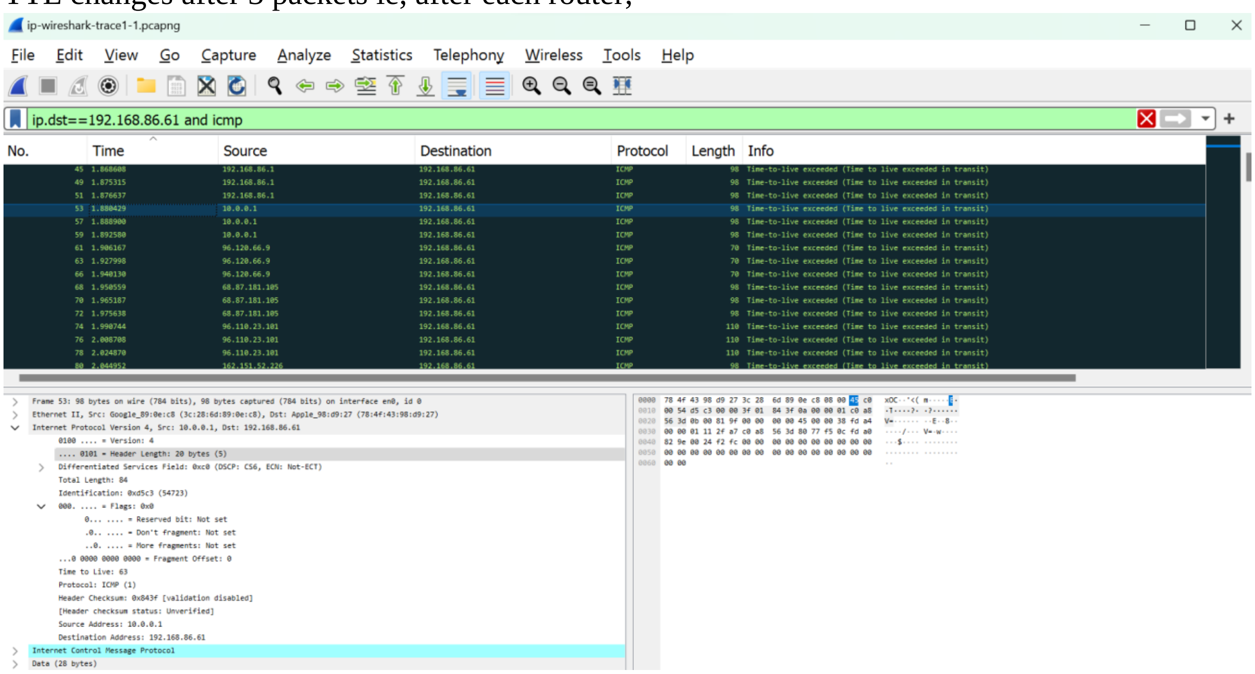
12. Are the values of the TTL fields similar, across all of ICMP packets from all of the routers?
ANSWER:
No, the values of TTL fields are not similar across all of ICMP packets from all of the routers. It is similar only for same 3 packets as screenshots attached below, but changes after every 3 packets. So, the values of TTL are not similar from all of the routers.

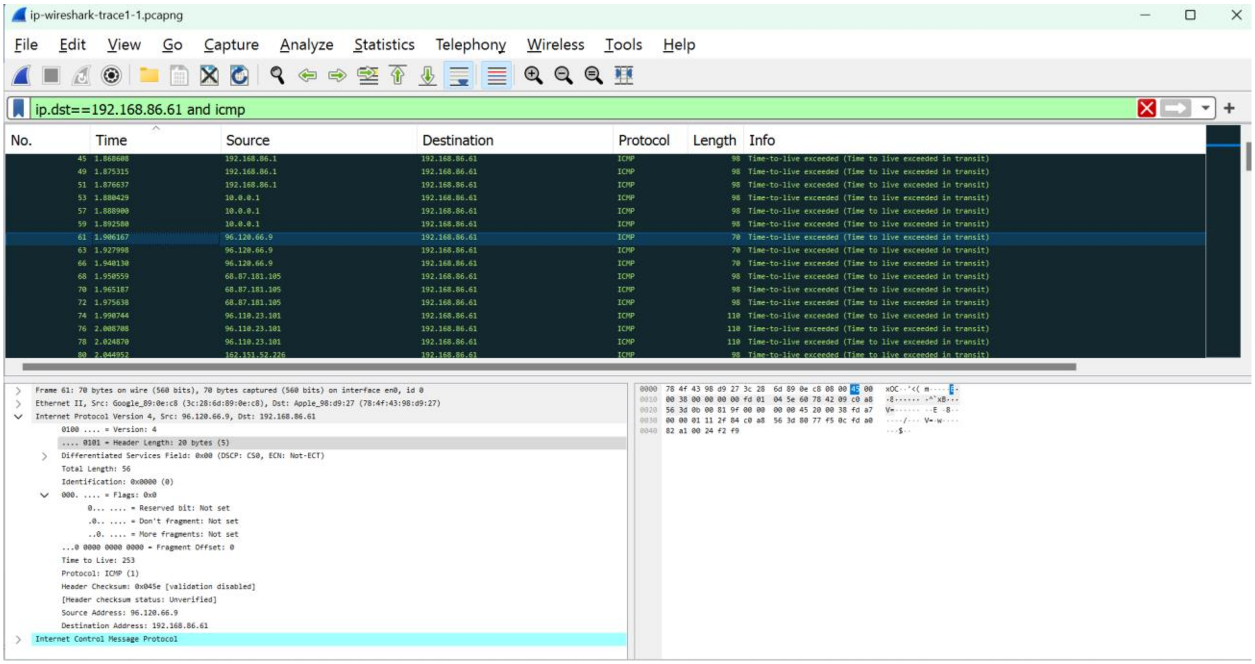

ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst==192.168.86.61 and icmp
No. Time Source Destination Protocol Length Info

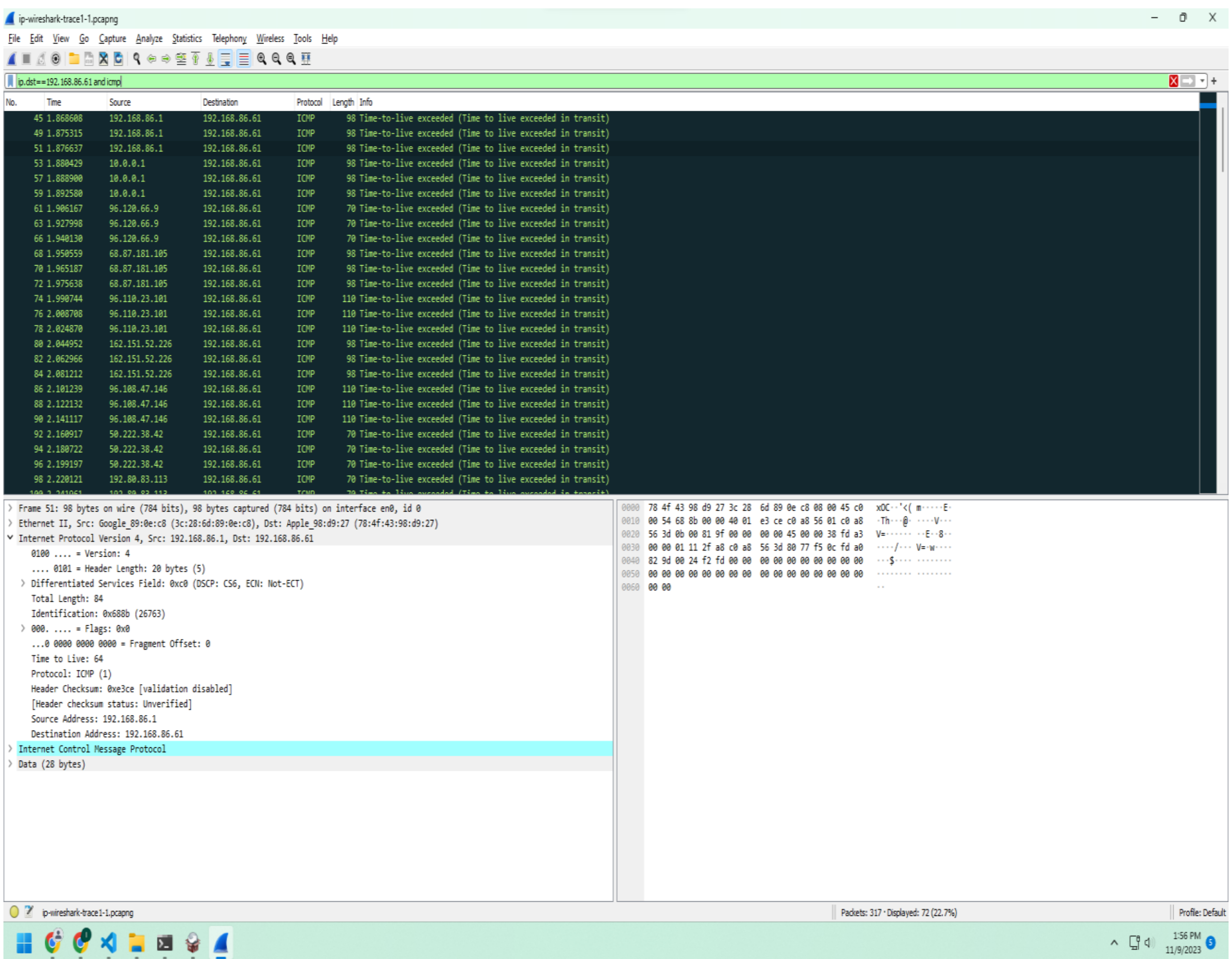

ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst==192.168.86.61 and icmp
No. Time Source Destination Protocol Length Info

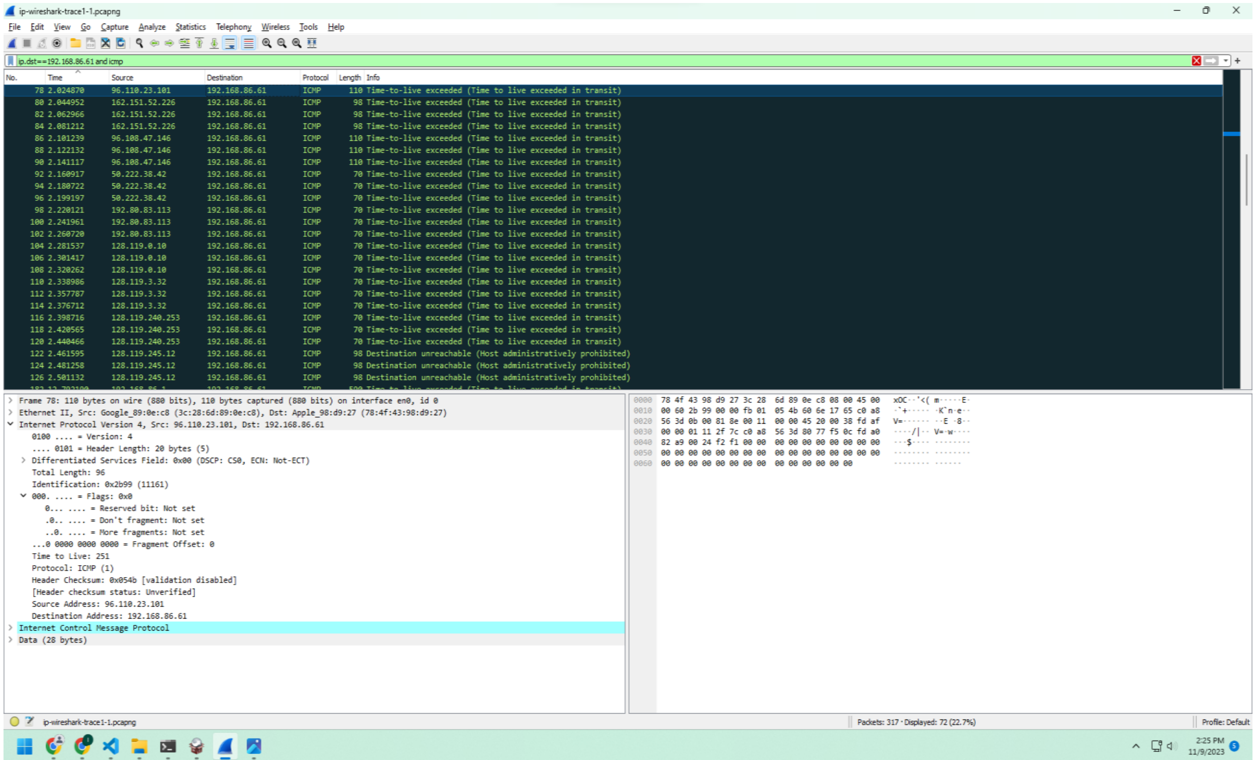


TTL changes after 3 packets ie, after each router,



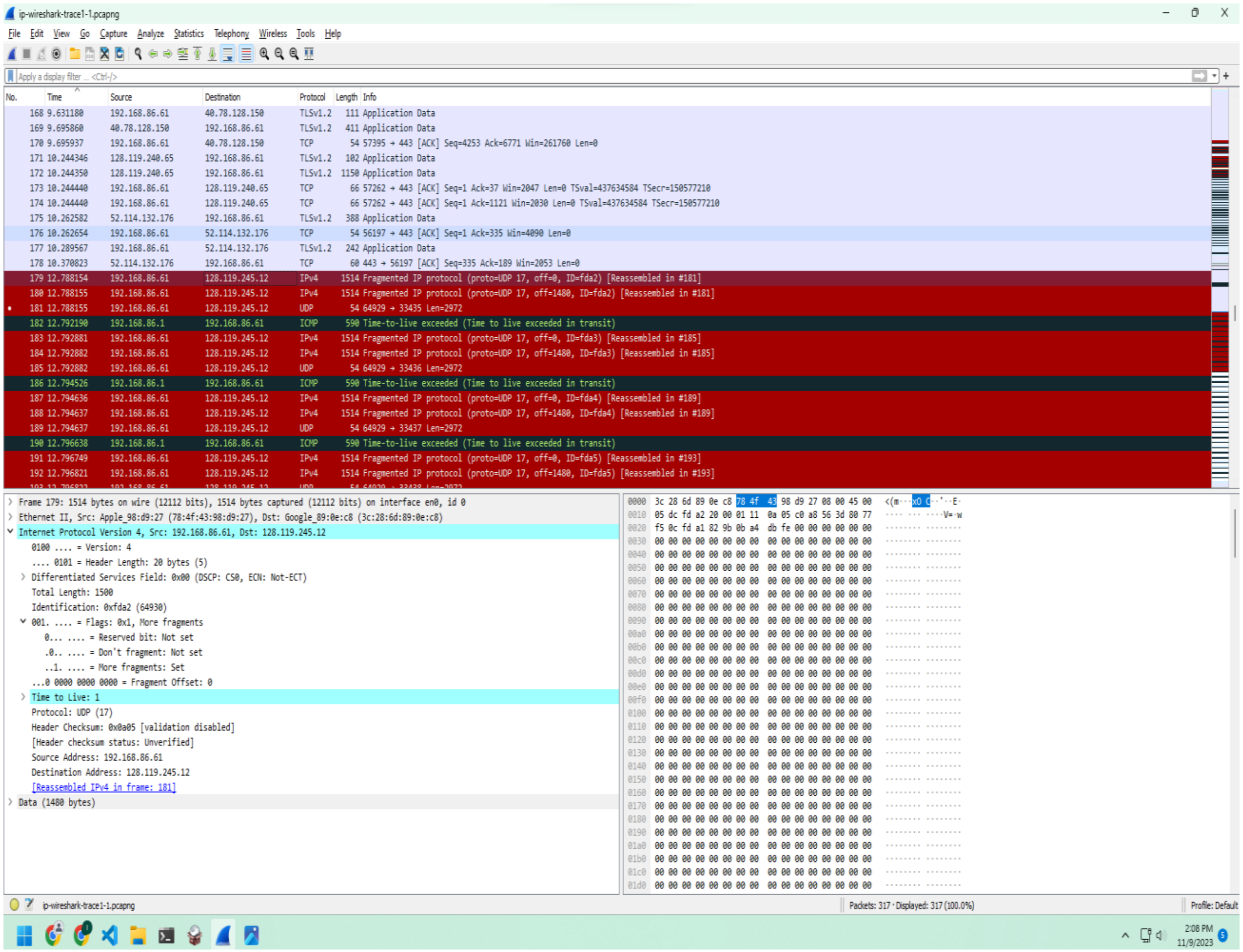

ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst==192.168.86.61 and icmp
No. Time Source Destination Protocol Length Info
Frame 61: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Google_89:0e:c8 (3c:28:6d:89:0e:c8), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
Internet Protocol Version 4, Src: 96.120.66.9, Dst: 192.168.86.61
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0x0000 (0)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 253
Protocol: ICMP (1)
Header Checksum: 0x045e [validation disabled]
[Header checksum status: Unverified]
Source Address: 96.120.66.9
Destination Address: 192.168.86.61
Internet Control Message Protocol


ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst==192.168.86.61 and icmp
No. Time Source Destination Protocol Length Info
45 1.868608 192.168.86.1 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
49 1.875315 192.168.86.1 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
51 1.876637 192.168.86.1 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
53 1.880429 10.0.0.1 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
57 1.888900 10.0.0.1 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
59 1.892580 10.0.0.1 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
61 1.906167 96.120.66.9 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
63 1.927998 96.120.66.9 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
66 1.940130 96.120.66.9 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
68 1.950559 68.87.181.105 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
70 1.965187 68.87.181.105 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
72 1.975638 68.87.181.105 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
74 1.990744 96.110.23.101 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
76 2.008708 96.110.23.101 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
78 2.024870 96.110.23.101 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
80 2.044952 162.151.52.226 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
82 2.062966 162.151.52.226 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
84 2.081212 162.151.52.226 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
86 2.101239 96.108.47.146 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
88 2.122132 96.108.47.146 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
90 2.141117 96.108.47.146 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
92 2.160917 50.222.38.42 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
94 2.180722 50.222.38.42 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
96 2.199197 50.222.38.42 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
98 2.220121 192.80.83.113 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
Frame 51: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
Ethernet II, Src: Google_89:0e:c8 (3c:28:6d:89:0e:c8), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
Internet Protocol Version 4, Src: 192.168.86.1, Dst: 192.168.86.61
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
Total Length: 84
Identification: 0x688b (26763)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0xe3ce [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.1
Destination Address: 192.168.86.61
Internet Control Message Protocol
Data (28 bytes)
ip-wireshark-trace1-1.pcapng
Packets: 317 · Displayed: 72 (22.7%)
Profile: Default
1:56 PM
11/9/2023



ip-wireshark-trace1-1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst==192.168.86.61 and icmp
No. Time Source Destination Protocol Length Info
78 2.024870 96.110.23.101 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
80 2.044952 162.151.52.226 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
82 2.062966 162.151.52.226 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
84 2.081212 162.151.52.226 192.168.86.61 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
86 2.101239 96.108.47.146 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
88 2.122132 96.108.47.146 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
90 2.141117 96.108.47.146 192.168.86.61 ICMP 110 Time-to-live exceeded (Time to live exceeded in transit)
92 2.160917 50.222.38.42 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
94 2.180722 50.222.38.42 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
96 2.199197 50.222.38.42 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
98 2.220121 192.80.83.113 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
100 2.241961 192.80.83.113 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
102 2.260720 192.80.83.113 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
104 2.281537 128.119.0.10 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
106 2.301417 128.119.0.10 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
108 2.320262 128.119.0.10 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
110 2.338986 128.119.3.32 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
112 2.357787 128.119.3.32 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
114 2.376712 128.119.3.32 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
116 2.398716 128.119.240.253 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
118 2.420565 128.119.240.253 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
120 2.440466 128.119.240.253 192.168.86.61 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
122 2.461595 128.119.245.12 192.168.86.61 ICMP 98 Destination unreachable (Host administratively prohibited)
124 2.481258 128.119.245.12 192.168.86.61 ICMP 98 Destination unreachable (Host administratively prohibited)
126 2.501132 128.119.245.12 192.168.86.61 ICMP 98 Destination unreachable (Host administratively prohibited)
Frame 78: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface en0, id 0
Ethernet II, Src: Google_89:0e:c8 (3c:28:6d:89:0e:c8), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
Internet Protocol Version 4, Src: 96.110.23.101, Dst: 192.168.86.61
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 96
Identification: 0x2b99 (11161)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 251
Protocol: ICMP (1)
Header Checksum: 0x054b [validation disabled]
[Header checksum status: Unverified]
Source Address: 96.110.23.101
Destination Address: 192.168.86.61
Internet Control Message Protocol
Data (28 bytes)
0000 78 4f 43 98 d9 27 3c 28 6d 89 0e c8 08 00 45 00
0010 00 60 2b 99 00 00 fb 01 05 4b 60 6e 17 65 c0 a8
0020 56 3d 0b 00 81 8e 00 11 00 00 45 20 00 38 fd af
0030 00 00 01 11 2f 7c c0 a8 56 3d 80 77 f5 0c fd a0
0040 82 a9 00 24 f2 f1 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00
ip-wireshark-trace1-1.pcapng
Packets: 317 · Displayed: 72 (22.7%)
Profile: Default
2:25 PM
11/9/2023

Part 2: Fragmentation



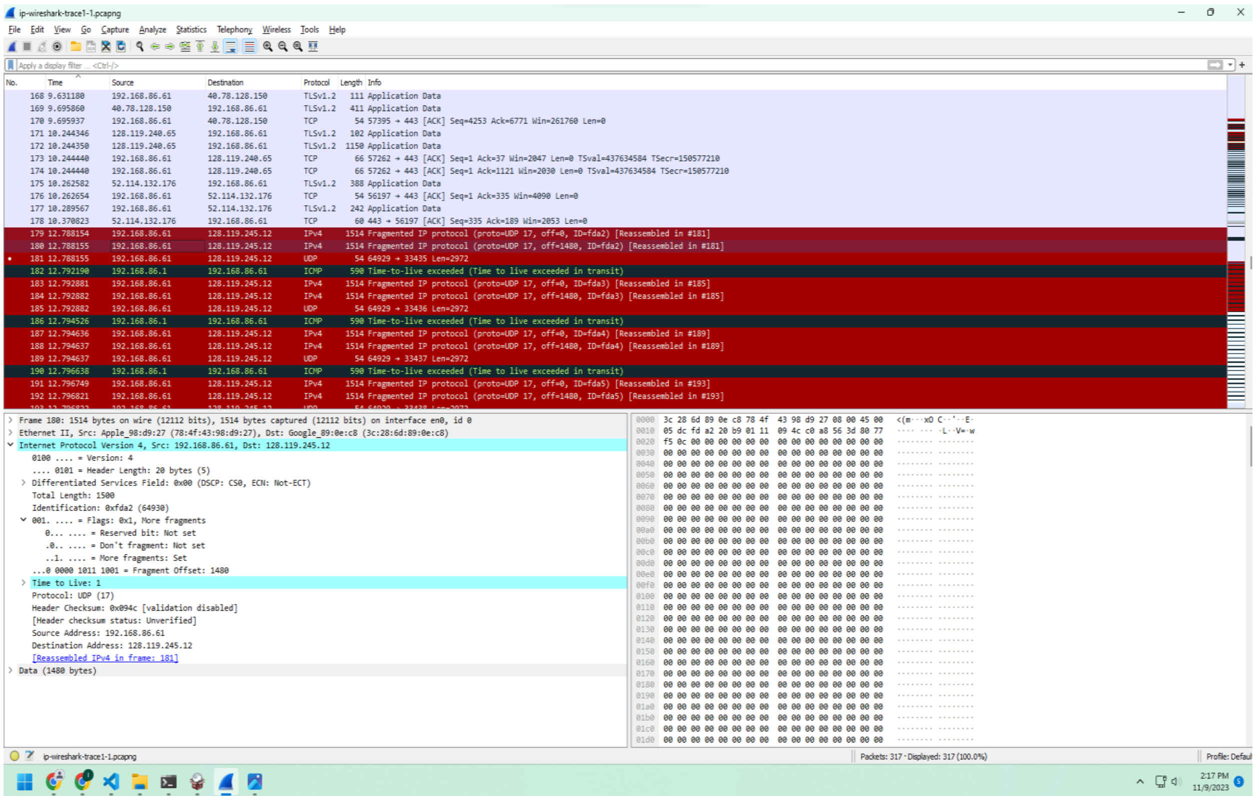
13. Find the first IP datagram containing the first part of the segment sent to 128.119.245.12 sent by your computer via the `traceroute` command to gaia.cs.umass.edu, *after* you specified that the `traceroute` packet length should be 3000. (Hint: This is packet 179 in the *ip-wireshark-trace1-1.pcapng* trace file.). Has that segment been fragmented across more than one IP datagram?
    ANSWER:
    Yes, packet 179 segment has been fragmented across more than one IP datagram.

14. What information in the IP header indicates that this datagram been fragmented?
    ANSWER:
    More fragment is set which indicates that this datagram has been fragmented.

15. What information in the IP header for this packet indicates whether this is the first fragment versus a latter fragment?

ANSWER:
Fragment offset: 0 indicates that packet 179 is 1st packet.

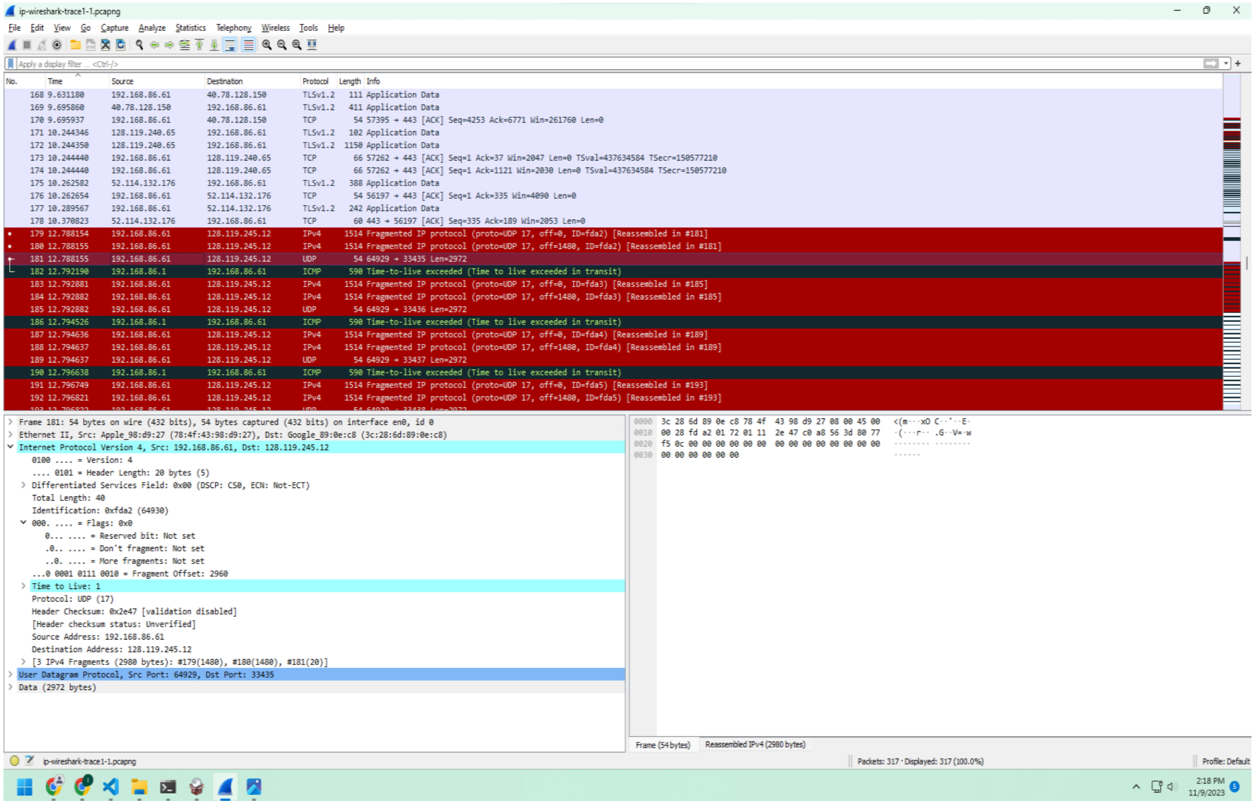
16. How many bytes are there in this IP datagram (header plus payload)?
ANSWER:
Total length: 1500bytes (header(20 bytes)+payload(1480 bytes)).



17. Now inspect the datagram containing the second fragment of the fragmented UDP segment. What information in the IP header indicates that this is *not* the first datagram fragment?
ANSWER:
Fragment offset: 1480 indicates that this is not 1st fragment.

18. What fields change in the IP header between the first and second fragment? List ALL the fields that change in the IP header.
ANSWER:
Fields change in the IP header between the first and second fragment

- Fragment offset
- Header Checksum



19. Now find the IP datagram containing the third fragment of the original UDP segment. What information in the IP header indicates that this is the last fragment of that segment?

    ANSWER:

    Fragment offset: 2960, so it is the 3rd fragment and More fragment is not set(0) indicates that this fragment is the last fragment.